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KGiSL INSTITUTE OF INFORMATION MANAGEMENT

Approved by *AICTE*, New Delhi and Affiliated to Bharathiar University

ISO 9001:2015 Certified Institution

KGiSL Campus, Saravanampatti, Coimbatore – 641 035

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Master of Computer Applications

Year 20 - 20

Department of Computer Science

KGiSL INSTITUTE OF INFORMATION MANAGEMENT

COIMBATORE-641 035

KGiSL INSTITUTE OF INFORMATION MANAGEMENT
COIMBATORE - 641 035
(Affiliated to Bharathiar University)

DEPARTMENT OF COMPUTER SCIENCE

LABORATORY RECORD

NAME: _____ **ROLL No.:** _____

SEMESTER: _____ **YEAR:** _____

SUBJECT: _____

Certified Bonafide Record of Work Done by

Mr / Ms _____

Staff In Charge HOD

Place : _____

Date : _____

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Internal Examiner External Examiner

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****** Program 1:Banking system*******

Primary Key, Unique Key creation

```
mysql> create table bank(accno int primary key, cust_name varchar(15),email varchar(100) unique,phonenumber varchar(15) unique);
Query OK, 0 rows affected (0.08 sec)
      desc bank;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| accno       | int         | NO   | PRI | NULL    |       |
| cust_name   | varchar(15) | YES  |     | NULL    |       |
| email       | varchar(100)| YES  | UNI | NULL    |       |
| phonenumber | varchar(15) | YES  | UNI | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)

mysql> insert into bank values(17071,'velan','velan01@gmail.com','9788934500');
Query OK, 1 row affected (0.02 sec)

mysql> insert into bank values(17072,'anu','anupriya@gmail.com','6734517000'),(17073,'shanthi','sanu21@gmail.com','7890012345');
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from bank;
+-------+----------+------------------+------------+
| accno | cust_name | email            | phonenumber |
+-------+----------+------------------+------------+
| 17071 | velan     | velan01@gmail.com  | 9788934500  |
| 17072 | anu       | anupriya@gmail.com | 6734517000  |
| 17073 | shanthi   | sanu21@gmail.com   | 7890012345  |
+-------+----------+------------------+------------+
3 rows in set (0.00 sec)
```

Unique key constraint Violation

```
mysql> insert into bank values (17074,'arun','anupriya@gmail.com','6734517000');
ERROR 1062 (23000): Duplicate entry 'anupriya@gmail.com' for key 'bank.email'
mysql>
```

Inserting NULL value in Unique key field

```
mysql> insert into bank values (17074,'arun',NULL,'9894238341');
Query OK, 1 row affected (0.02 sec)

mysql> insert into bank values (17075,'padma','padma@gmail.com',null);
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into bank values (17076,'preethi',null,null);
Query OK, 1 row affected (0.02 sec)

mysql> select * from bank;
+-------+-----------+--------------------+-------------+
| accno | cust_name | email              | phonenumber |
+-------+-----------+--------------------+-------------+
| 17071 | velan     | velan01@gmail.com  | 9788934500  |
| 17072 | anu       | anupriya@gmail.com | 6734517000  |
| 17073 | shanthi   | sanu21@gmail.com   | 7890012345  |
| 17074 | arun      | NULL               | 9894238341  |
| 17075 | padma     | padma@gmail.com    | NULL        |
| 17076 | preethi   | NULL               | NULL        |
+-------+-----------+--------------------+-------------+
6 rows in set (0.00 sec)
```

Foreign Key creation

```
mysql> create table loandetails(loanid int primary key,accno int,loanamt int, foreign key (accno) references bank(accno));
Query OK, 0 rows affected (0.04 sec)

mysql> desc loandetails;
+---------+------+------+-----+---------+-------+
| Field   | Type | Null | Key | Default | Extra |
+---------+------+------+-----+---------+-------+
| loanid  | int  | NO   | PRI | NULL    |       |
| accno   | int  | YES  | MUL | NULL    |       |
| loanamt | int  | YES  |     | NULL    |       |
+---------+------+------+-----+---------+-------+
3 rows in set (0.01 sec)

mysql> insert into loandetails values(101,17071,50000);
Query OK, 1 row affected (0.02 sec)

mysql> insert into loandetails values(102,17074,60000);
Query OK, 1 row affected (0.02 sec)

mysql> insert into loandetails values(103,17077,45000);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`db1`.`loandetails`, CONSTRAINT `loandetails_ib
fk_1` FOREIGN KEY (`accno`) REFERENCES `bank` (`accno`))
mysql> select * from loandetails;
+--------+-------+---------+
| loanid | accno | loanamt |
+--------+-------+---------+
|    101 | 17071 |   50000 |
|    102 | 17074 |   60000 |
+--------+-------+---------+
2 rows in set (0.00 sec)
```

Combining two table data using Joining function

```
mysql> select b.cust_name,l.loanid,l.loanamt from bank b inner join loandetails as l on b.accno=l.accno;
+-----------+--------+----------+
| cust_name | loanid | loanamt |
+-----------+--------+----------+
| velan     |    101 |   50000 |
| arun      |    102 |   60000 |
+-----------+--------+----------+
2 rows in set (0.00 sec)

mysql>
```

****

Program2: *****Online Reservation System*****

Table1: Table creation, value insertion & viewing data and its field.

Table Name: Customer **Primary key:** c_id

```
mysql> create table customer (c_id int primary key,c_name varchar(10),email varchar(25),phonenumber varchar(10));
Query OK, 0 rows affected (0.04 sec)

mysql> insert into customer values(1071,'sethu','sethu@gmail.com','7634529800');
Query OK, 1 row affected (0.02 sec)

mysql> insert into customer values(1072,'mani','mani@gmail.com','9843978876'),(1073,'jancy','jancy15@gmail.com','9342156567');
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from customer;
+------+--------+------------------+-------------+
| c_id | c_name | email            | phonenumber |
+------+--------+------------------+-------------+
| 1071 | sethu  | sethu@gmail.com  | 7634529800  |
| 1072 | mani   | mani@gmail.com   | 9843978876  |
| 1073 | jancy  | jancy15@gmail.com | 9342156567  |
+------+--------+------------------+-------------+
3 rows in set (0.00 sec)

mysql> desc customer;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| c_id        | int         | NO   | PRI | NULL    |       |
| c_name      | varchar(10) | YES  |     | NULL    |       |
| email       | varchar(25) | YES  |     | NULL    |       |
| phonenumber | varchar(10) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Table2: Creation & description of the table

Table Name: Vehicle **Primary key:** vid

```
mysql> create table vehicle(vid int primary key,v_type enum('car','van','bus'),capacity int);
Query OK, 0 rows affected (0.04 sec)

mysql> desc vehicle;
+----------+-------------------------+------+-----+---------+-------+
| Field    | Type                    | Null | Key | Default | Extra |
+----------+-------------------------+------+-----+---------+-------+
| vid      | int                     | NO   | PRI | NULL    |       |
| v_type   | enum('car','van','bus') | YES  |     | NULL    |       |
| capacity | int                     | YES  |     | NULL    |       |
+----------+-------------------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

Value Insertion and enum data type setting data violation

```
mysql> insert into vehicle values(1001,'car',4);
Query OK, 1 row affected (0.02 sec)

mysql> insert into vehicle values(1200,'van',10);
Query OK, 1 row affected (0.02 sec)

mysql>  insert into vehicle values(1201,'bus',52);
Query OK, 1 row affected (0.01 sec)

mysql> insert into vehicle values(1201,'bike',1);
ERROR 1265 (01000): Data truncated for column 'v_type' at row 1
```

Viewing vehicle table data

```
mysql> select * from vehicle;
+------+--------+----------+
| vid  | v_type | capacity |
+------+--------+----------+
| 1001 | car    |        4 |
| 1200 | van    |       10 |
| 1201 | bus    |       52 |
+------+--------+----------+
3 rows in set (0.00 sec)
```

Table3: creation & setting check integrity to the field start_datetime

Table Name: reserve **Primary key:**r_id **Foreign key:** c_id,vid

Check Integrity Constraint assigned to start_datetime column

```
mysql> create table reserve(r_id int primary key,c_id int,vid int,start_datetime datetime check(start_datetime >='2025-10-25'),end_da
tetime datetime,palce varchar(15),foreign key(c_id) references customer(c_id),foreign key(vid) references vehicle(vid));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> desc reserve;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| r_id           | int         | NO   | PRI | NULL    |       |
| c_id           | int         | YES  | MUL | NULL    |       |
| vid            | int         | YES  | MUL | NULL    |       |
| start_datetime | datetime    | YES  |     | NULL    |       |
| end_datetime   | datetime    | YES  |     | NULL    |       |
| place          | varchar(15) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

```
mysql> insert into reserve values(1,1071,1200,'2025-10-26 06:00:00','2025-10-26 09:00:00','salem');
Query OK, 1 row affected (0.02 sec)

mysql> insert into reserve values(2,1073,1201,'2025-10-25 10:30:00','2025-10-27 06:00:00','delhi');
Query OK, 1 row affected (0.02 sec)

mysql>  insert into reserve values(3,1072,1200,'2025-10-20 10:00:00','2025-10-25 08:00:00','madurai');
ERROR 3819 (HY000): Check constraint 'reserve_chk_1' is violated.
mysql> select * from reserve;
+------+------+------+---------------------+---------------------+-------+
| r_id | c_id | vid  | start_datetime      | end_datetime        | place |
+------+------+------+---------------------+---------------------+-------+
|    1 | 1071 | 1200 | 2025-10-26 06:00:00 | 2025-10-26 09:00:00 | salem |
|    2 | 1073 | 1201 | 2025-10-25 10:30:00 | 2025-10-27 06:00:00 | delhi |
+------+------+------+---------------------+---------------------+-------+
2 rows in set (0.00 sec)
```

****

Program3: *****Personal Information*****

```
mysql> create table personal_data(username varchar(10),emailid varchar(25) unique,UG_University varchar(25));
Query OK, 0 rows affected (0.08 sec)

mysql> desc personal_data;
+--------------+------------+------+-----+---------+-------+
| Field        | Type       | Null | Key | Default | Extra |
+--------------+------------+------+-----+---------+-------+
| username     | varchar(10) | YES  |     | NULL    |       |
| emailid      | varchar(25) | YES  | UNI | NULL    |       |
| UG_University | varchar(25) | YES  |     | NULL    |       |
+--------------+------------+------+-----+---------+-------+
3 rows in set (0.02 sec)

mysql> insert into personal_data values('sumathi','sumathi15@gmail.com','bharathiyar'),('ravi','ravi23@gmail.com','bhara
thidasan'),('keerthi','keerthi030@gmail.com','bharathiar');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> insert into personal_data values('sumathi','suma2104@gmail.com','periyar');
Query OK, 1 row affected (0.00 sec)

mysql> insert into personal_data values('kumar','kumar101@gmail.com','bharathiar');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select * from personal_data;
+----------+---------------------+---------------+
| username | emailid             | UG_University |
+----------+---------------------+---------------+
| sumathi  | sumathi15@gmail.com | bharathiyar   |
| ravi     | ravi23@gmail.com    | bharathidasan |
| keerthi  | keerthi030@gmail.com | bharathiar    |
| sumathi  | suma2104@gmail.com  | periyar       |
| kumar    | kumar101@gmail.com  | bharathiar    |
+----------+---------------------+---------------+
5 rows in set (0.00 sec)
```

Using LIKE Operator

```
mysql> SELECT * FROM personal_data WHERE emailid  LIKE '%@gmail.com%';
+----------+----------------------+---------------+
| username | emailid              | UG_University |
+----------+----------------------+---------------+
| sumathi  | sumathi15@gmail.com  | bharathiyar   |
| ravi     | ravi23@gmail.com     | bharathidasan |
| keerthi  | keerthi030@gmail.com | bharathiar    |
| sumathi  | suma2104@gmail.com   | periyar       |
| kumar    | kumar101@gmail.com   | bharathiar    |
+----------+----------------------+---------------+
5 rows in set (0.01 sec)

mysql> SELECT * FROM personal_data WHERE username LIKE '_a%';
+----------+------------------+---------------+
| username | emailid          | UG_University |
+----------+------------------+---------------+
| ravi     | ravi23@gmail.com | bharathidasan |
+----------+------------------+---------------+
1 row in set (0.00 sec)

mysql>  SELECT * FROM personal_data WHERE username LIKE '%a_';
+----------+--------------------+---------------+
| username | emailid            | UG_University |
+----------+--------------------+---------------+
| kumar    | kumar101@gmail.com | bharathiar    |
+----------+--------------------+---------------+
1 row in set (0.00 sec)
```

```
mysql> SELECT * FROM personal_data WHERE UG_University LIKE '%a_';
+----------+----------------------+---------------+
| username | emailid              | UG_University |
+----------+----------------------+---------------+
| sumathi  | sumathi15@gmail.com  | bharathiyar   |
| ravi     | ravi23@gmail.com     | bharathidasan |
| keerthi  | keerthi030@gmail.com | bharathiar    |
| sumathi  | suma2104@gmail.com   | periyar       |
| kumar    | kumar101@gmail.com   | bharathiar    |
+----------+----------------------+---------------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM personal_data WHERE UG_University LIKE '%ar';
+----------+----------------------+---------------+
| username | emailid              | UG_University |
+----------+----------------------+---------------+
| sumathi  | sumathi15@gmail.com  | bharathiyar   |
| keerthi  | keerthi030@gmail.com | bharathiar    |
| sumathi  | suma2104@gmail.com   | periyar       |
| kumar    | kumar101@gmail.com   | bharathiar    |
+----------+----------------------+---------------+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM personal_data WHERE UG_University NOT LIKE '_h%';
+----------+--------------------+---------------+
| username | emailid            | UG_University |
+----------+--------------------+---------------+
| sumathi  | suma2104@gmail.com | periyar       |
+----------+--------------------+---------------+
1 row in set (0.01 sec)
```

Group by with Aggregate functions in USER Table

```
mysql> select * from user;
+--------+----------+-------+----------+
| Iduser | username | place | invested |
+--------+----------+-------+----------+
|   1001 | anu      | cbe   |     NULL |
|   1002 | priya    | ooty  |    10000 |
|   1003 | monika   | cbe   |    10000 |
|   1005 | kiru     | ooty  |    10000 |
+--------+----------+-------+----------+
4 rows in set (0.00 sec)

mysql> select count(*) from user;
+----------+
| count(*) |
+----------+
|        4 |
+----------+
1 row in set (0.02 sec)

mysql> select sum(invested) from user;
+---------------+
| sum(invested) |
+---------------+
|         30000 |
+---------------+
1 row in set (0.00 sec)

mysql> select avg(invested) from user;
+---------------+
| avg(invested) |
+---------------+
|    10000.0000 |
+---------------+
```

```
mysql> update user set invested=5000 where Iduser=1003;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from user;
+--------+----------+-------+----------+
| Iduser | username | place | invested |
+--------+----------+-------+----------+
|   1001 | anu      | cbe   |     NULL |
|   1002 | priya    | ooty  |    10000 |
|   1003 | monika   | cbe   |     5000 |
|   1005 | kiru     | ooty  |    15000 |
+--------+----------+-------+----------+
4 rows in set (0.02 sec)
```

```
mysql>  select place, avg(invested) from user group by place;
+-------+---------------+
| place | avg(invested) |
+-------+---------------+
| cbe   |     5000.0000 |
| ooty  |    10000.0000 |
+-------+---------------+
2 rows in set (0.02 sec)
```

```
mysql> select place,min(invested) from user group by place;
+-------+---------------+
| place | min(invested) |
+-------+---------------+
| cbe   |          5000 |
| ooty  |         10000 |
+-------+---------------+
2 rows in set (0.01 sec)

mysql> select place,sum(invested) from user group by place;
+-------+---------------+
| place | sum(invested) |
+-------+---------------+
| cbe   |          5000 |
| ooty  |         25000 |
+-------+---------------+
2 rows in set (0.00 sec)
```

Order by and Limit

```
mysql> select * from personalinfo order by studentname asc;
+---------+-------------+-------------+------------+--------------------+---------------------+
| rollno  | studentname | fathername  | occupation | email              | created_at          |
+---------+-------------+-------------+------------+--------------------+---------------------+
| 25MCA04 | Abi         | Arunachalam | Manager    | abi01@gamil.com    | 2025-09-02 14:47:13 |
| 25MCA01 | Anu         | Murugan     | Manager    | anupriya@gmail.com | 2025-09-02 14:17:57 |
| 25MCA02 | Banupriya   | Balaji      | salesman   | banu@gmail.com     | 2025-09-02 14:19:55 |
| 25MCA06 | Mani        | Senthil     | Teacher    | mani007@gmail.com  | 2025-09-02 14:55:33 |
| 25MCA03 | Prakash     | Kumar       | Clerk      | prakash21@gmail.com | 2025-09-02 14:21:00 |
+---------+-------------+-------------+------------+--------------------+---------------------+
5 rows in set (0.00 sec)

mysql> select * from personalinfo order by studentname asc limit 1;
+---------+-------------+-------------+------------+----------------+---------------------+
| rollno  | studentname | fathername  | occupation | email          | created_at          |
+---------+-------------+-------------+------------+----------------+---------------------+
| 25MCA04 | Abi         | Arunachalam | Manager    | abi01@gamil.com | 2025-09-02 14:47:13 |
+---------+-------------+-------------+------------+----------------+---------------------+
1 row in set (0.00 sec)

mysql> select * from personalinfo order by studentname desc limit 1;
+---------+-------------+-----------+-----------+--------------------+---------------------+
| rollno  | studentname | fathername | occupation | email              | created_at          |
+---------+-------------+-----------+-----------+--------------------+---------------------+
| 25MCA03 | Prakash     | Kumar     | Clerk      | prakash21@gmail.com | 2025-09-02 14:21:00 |
+---------+-------------+-----------+-----------+--------------------+---------------------+
1 row in set (0.00 sec)
```

```
mysql> select * from emp;
+------+--------+------+------+
| eid  | ename  | age  | edu  |
+------+--------+------+------+
|    1 | sindhu |   25 | mca  |
|    2 | kavi   |   26 | mca  |
+------+--------+------+------+
2 rows in set (0.02 sec)

mysql> insert into emp values(3,'rithu',25,'mca'),(4,'raju',26,'mca'),(5,'ganesh',25,'mca'),(6,'priya',26,'mca');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> select * from emp;
+------+--------+------+------+
| eid  | ename  | age  | edu  |
+------+--------+------+------+
|    1 | sindhu |   25 | mca  |
|    2 | kavi   |   26 | mca  |
|    3 | rithu  |   25 | mca  |
|    4 | raju   |   26 | mca  |
|    5 | ganesh |   25 | mca  |
|    6 | priya  |   26 | mca  |
+------+--------+------+------+
6 rows in set (0.00 sec)
```

Having Clause in EMP Table

```
mysql> select edu,count(eid)as num_employees from emp group by edu having count(eid)>3;
+------+---------------+
| edu  | num_employees |
+------+---------------+
| mca  |             6 |
+------+---------------+
1 row in set (0.02 sec)

mysql> select age,count(eid)as num_employees from emp group by age having count(eid)>=3;
+------+---------------+
| age  | num_employees |
+------+---------------+
|   25 |             3 |
|   26 |             3 |
+------+---------------+
2 rows in set (0.00 sec)
```

Program 4: ***** Student mark processing system*****

Table creation

```
mysql> create table studentdata(rollno varchar(8) primary key,studentname varchar(15),java int,adbms int,cn int,os int,d_net int,tota
l int,percentage int);
Query OK, 0 rows affected (0.04 sec)

mysql> desc studentdata;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| rollno      | varchar(8)  | NO   | PRI | NULL    |       |
| studentname | varchar(15) | YES  |     | NULL    |       |
| java        | int         | YES  |     | NULL    |       |
| adbms       | int         | YES  |     | NULL    |       |
| cn          | int         | YES  |     | NULL    |       |
| os          | int         | YES  |     | NULL    |       |
| d_net       | int         | YES  |     | NULL    |       |
| total       | int         | YES  |     | NULL    |       |
| percentage  | int         | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
9 rows in set (0.01 sec)
```

Insert Values to Table

```
mysql> insert into studentdata (rollno,studentname,java,adbms,cn,os,d_net)values('25mca01','arun',50,57,67,78,50);
Query OK, 1 row affected (0.02 sec)

mysql> insert into studentdata (rollno,studentname,java,adbms,cn,os,d_net)values('25mca02','banu',65,57,69,60,48),('25mca03','chitra'
,78,67,56,46,98);
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from studentdata;
+---------+-------------+------+-------+------+------+-------+-------+------------+
| rollno  | studentname | java | adbms | cn   | os   | d_net | total | percentage |
+---------+-------------+------+-------+------+------+-------+-------+------------+
| 25mca01 | arun        |   50 |    57 |   67 |   78 |    50 |  NULL |       NULL |
| 25mca02 | banu        |   65 |    57 |   69 |   60 |    48 |  NULL |       NULL |
| 25mca03 | chitra      |   78 |    67 |   56 |   46 |    98 |  NULL |       NULL |
+---------+-------------+------+-------+------+------+-------+-------+------------+
3 rows in set (0.00 sec)
```

Update total, percentage column

```
mysql> update studentdata set total=java+adbms+cn+os+d_net;
Query OK, 3 rows affected (0.02 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> update studentdata set percentage=total/5;
Query OK, 3 rows affected (0.02 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> select * from studentdata;
+---------+-------------+------+-------+------+------+-------+-------+------------+
| rollno  | studentname | java | adbms | cn   | os   | d_net | total | percentage |
+---------+-------------+------+-------+------+------+-------+-------+------------+
| 25mca01 | arun        |   50 |    57 |   67 |   78 |    50 |   302 |         60 |
| 25mca02 | banu        |   65 |    57 |   69 |   60 |    48 |   299 |         60 |
| 25mca03 | chitra      |   78 |    67 |   56 |   46 |    98 |   345 |         69 |
+---------+-------------+------+-------+------+------+-------+-------+------------+
3 rows in set (0.00 sec)
```

Logical Operator using Where Clause

```
mysql> select rollno,java from studentdata where java between 45 and 65;
+---------+------+
| rollno  | java |
+---------+------+
| 25mca01 |   50 |
| 25mca02 |   65 |
+---------+------+
2 rows in set (0.00 sec)

mysql> select rollno,java,d_net from studentdata where java>45 and d_net>45;
+---------+------+-------+
| rollno  | java | d_net |
+---------+------+-------+
| 25mca01 |   50 |    50 |
| 25mca02 |   65 |    48 |
| 25mca03 |   78 |    98 |
+---------+------+-------+
3 rows in set (0.00 sec)

mysql> select rollno,cn,os from studentdata where cn>=60 or os>=60;
+---------+------+------+
| rollno  | cn   | os   |
+---------+------+------+
| 25mca01 |   67 |   78 |
| 25mca02 |   69 |   60 |
+---------+------+------+
2 rows in set (0.00 sec)
```

```
mysql>  select rollno,studentname from studentdata where total not between 300 and 350;
+---------+-------------+
| rollno  | studentname |
+---------+-------------+
| 25mca02 | banu        |
+---------+-------------+
1 row in set (0.00 sec)

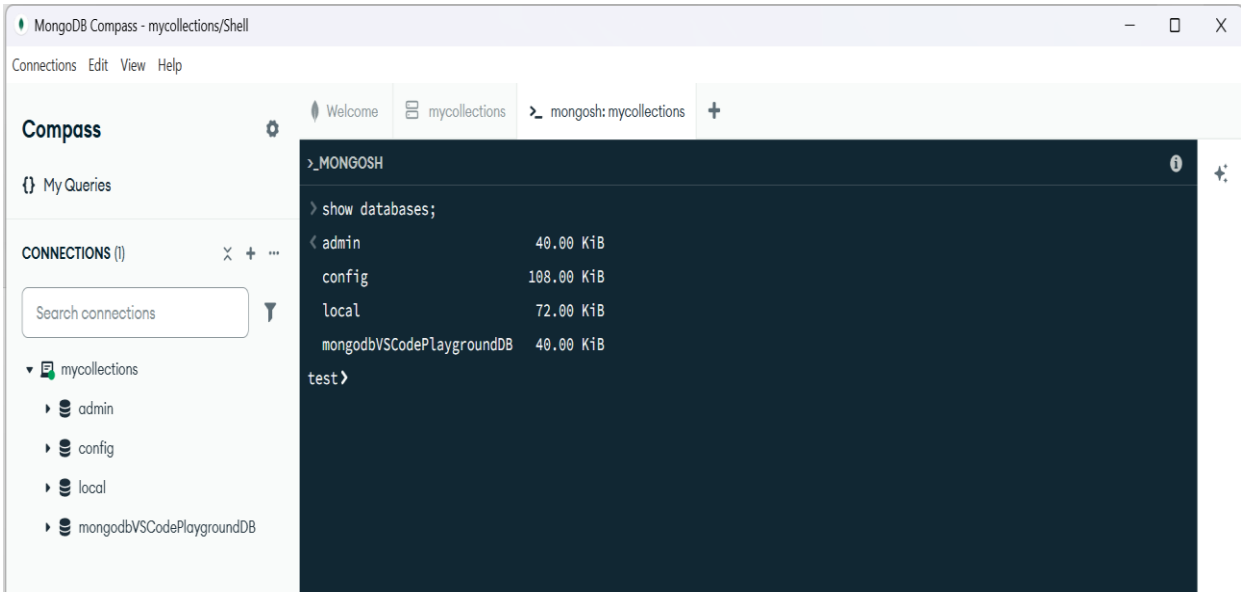
mysql> select rollno,studentname from studentdata where total <> 302;
+---------+-------------+
| rollno  | studentname |
+---------+-------------+
| 25mca02 | banu        |
| 25mca03 | chitra      |
+---------+-------------+
2 rows in set (0.00 sec)

mysql> select rollno,studentname,total from studentdata where total=299;
+---------+-------------+-------+
| rollno  | studentname | total |
+---------+-------------+-------+
| 25mca02 | banu        |   299 |
+---------+-------------+-------+
1 row in set (0.00 sec)
```

****

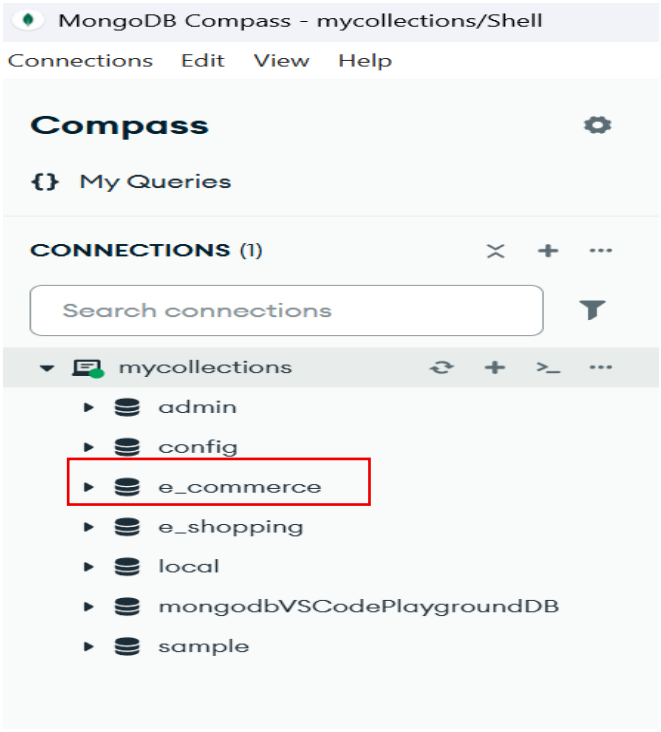
*******Program 5: Design an E-Commerce product catalog system using MongoDB as a storage engine and insert values. ********

Step 1: Open mongoshell:



Step 2: show databases; and shift to newly created database named as e_commerce by using query use e_commere.

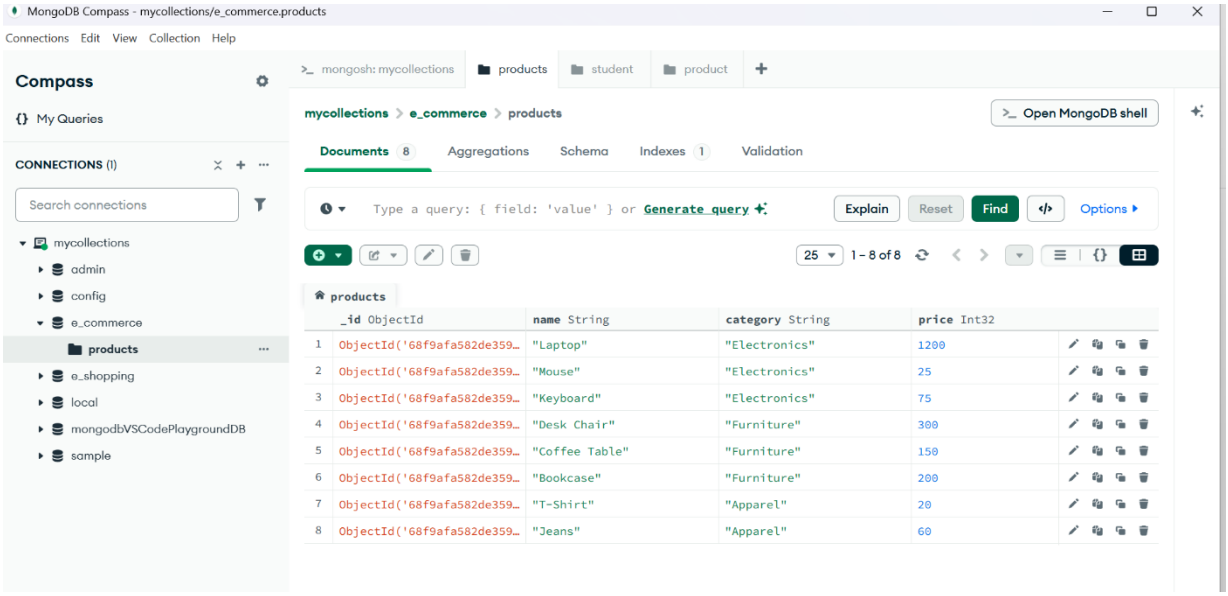
```
> use e_commerce
< switched to db e_commerce
```



Step 3: Insert values to the new collection name as product and view the created collections

```
> db.products.insertMany([
    { name: "Laptop", category: "Electronics", price: 1200, stock: 50 },
    { name: "Mouse", category: "Electronics", price: 25, stock: 200 },
    { name: "Keyboard", category: "Electronics", price: 75, stock: 150 },
    { name: "Desk Chair", category: "Furniture", price: 300, stock: 30 },
    { name: "Coffee Table", category: "Furniture", price: 150, stock: 40 },
    { name: "Bookcase", category: "Furniture", price: 200, stock: 20 },
    { name: "T-Shirt", category: "Apparel", price: 20, stock: 500 },
    { name: "Jeans", category: "Apparel", price: 60, stock: 100 }
  ]);
```

Step 4: View the data in Table view



Step 5: View data in mongoshell

```
>_MONGOSH
> db.products.find()
< {
    _id: ObjectId('68f9afa582de359b62abe9ef'),
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    stock: 50
  }
  {
    _id: ObjectId('68f9afa582de359b62abe9f0'),
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    stock: 200
  }
  {
    _id: ObjectId('68f9afa582de359b62abe9f1'),
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    stock: 150
  }
```

Program 6: Perform basic CURD operations (Create, Update, Read and Delete) functions for the product catalog.

Step 1: Create collection **collegeshop** in database **e_shopping**

```
> use e_shopping
< switched to db e_shopping
> db.createCollection("collegeshop")
< { ok: 1 }
```

Step 2: Insert one document in **collegeshop**

```
> db.collegeshop.insertOne({item:"notebook",qty:50})
< {
    acknowledged: true,
    insertedId: ObjectId('68e72b45771f57bccf5daf3b')
  }
```

Insert many documents in **collegeshop**

```
> db.collegeshop.insertMany([{item:"ballpointpen",qty:100},{item:"inkpen",qty:60}])
< {
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('68e72b8e771f57bccf5daf3c'),
      '1': ObjectId('68e72b8e771f57bccf5daf3d')
    }
  }
```

Step 3: Read documents from collection using **find()**

```
> db.collegeshop.find({qty:{$gt : 15 }})
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60
  }
```

Read documents from collection using **findOne()**

```
> db.collegeshop.findOne({item:"ballpointpen"})
< {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100
  }
```

Step 4: To find documents where **qty** is greater than **30** and less than **100**.

```
> db.collegeshop.find({ qty: { $gt: 30, $lt: 100 } })
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60
  }
```

To find documents where **qty** is greater than or equal to 60 and less than or equal to**100**.

```
> db.collegeshop.find({ qty: { $gte: 60, $lte: 100 } })
< {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60
  }
```

Step 5: To **Update** one document from the collection **collegeshop**

```
> db.collegeshop.updateOne(
    { item: "notebook" },
    { $set: { "category": "paperstuff" } }
  )
```

To view the data updation use **find()**

```
> db.collegeshop.find();
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50,
    category: 'paperstuff'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60
  }
```

Updating other two document one by one using **updateone**

```
> db.collegeshop.updateOne(
    { item: "ballpointpen" },
    { $set: { "category": "writingstuff" } }
  )
```

```
> db.collegeshop.updateOne(
    { item: "inkpen" },
    { $set: { "category": "writingstuff" } }
  )
```

Step 6: To UpdateMany document one at a time (**all documents that match the specified filter criteria**)

```
> db.collegeshop.updateMany({category:"writingstuff"},{$set:{satus:"available"}})
```

```
> db.collegeshop.updatemany({category:"writingstuff"},{$set:{satus:"available"}})
TypeError: db.collegeshop.updatemany is not a function
```

```
> db.collegeshop.find();
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50,
    category: 'paperstuff'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60
  }
```

Proceed step 5 to update the document and continue with step 6 then output

```
> db.collegeshop.find()
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50,
    category: 'paperstuff'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100,
    category: 'writingstuff'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60,
    category: 'writingstuff'
  }
```

```
> db.collegeshop.updateMany({category:"writingstuff"},{$set:{satus:"available"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 2,
    modifiedCount: 2,
    upsertedCount: 0
  }
```

```
> db.collegeshop.find()
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50,
    category: 'paperstuff'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100,
    category: 'writingstuff',
    satus: 'available'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'inkpen',
    qty: 60,
    category: 'writingstuff',
    satus: 'available'
  }
```

Step 7: To replaceone this document with a new one:

```
> db.collegeshop.replaceOne({item:"inkpen"},{item:"blueinkpen",qty:70,category:"writingstuff",status:"available"})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

The item inkpen data is replaced by using above query

```
> db.collegeshop.find()
< {
    _id: ObjectId('68e72b45771f57bccf5daf3b'),
    item: 'notebook',
    qty: 50,
    category: 'paperstuff'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100,
    category: 'writingstuff',
    satus: 'available'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'blueinkpen',
    qty: 70,
    category: 'writingstuff',
    status: 'available'
  }
```

```
> db.collegeshop.insertOne({item:"colourbox",qty:55,category:"writingstuff",status:"available"})
< {
    acknowledged: true,
    insertedId: ObjectId('68e74dff771f57bccf5daf3e')
  }
```

```
> db.collegeshop.find()
< {
    _id: ObjectId('68e72b8e771f57bccf5daf3c'),
    item: 'ballpointpen',
    qty: 100,
    category: 'writingstuff',
    satus: 'available'
  }
  {
    _id: ObjectId('68e72b8e771f57bccf5daf3d'),
    item: 'blueinkpen',
    qty: 70,
    category: 'writingstuff',
    status: 'available'
  }
  {
    _id: ObjectId('68e74dff771f57bccf5daf3e'),
    item: 'colourbox',
    qty: 55,
    category: 'writingstuff',
```

Deleting a Single Document:

```
> db.collegeshop.deleteOne({item:"colourbox"})
< {
    acknowledged: true,
    deletedCount: 1
  }
```

Deleting Many Document:

```
> db.collegeshop.deleteMany({category:"writingstuff"})
< {
    acknowledged: true,
    deletedCount: 2
  }
> db.collegeshop.find()
<
```

******Program 7: Built-in function in MongoDB*****

Step 1 : Use e_shopping database → insert values to s_mark collection

```
> use e_shopping
< switched to db e_shopping
```

```
> db.s_mark.insertOne({_id:1,sname:"Arun",java:76,cn:73,os:98});
< {
    acknowledged: true,
    insertedId: 1
  }
```

```
> db.s_mark.insertMany([{_id:2,sname:"banu",java:67,cn:56,os:76},{_id:3,sname:"chitra",java:89,cn:67,os:54},
{_id:4,sname:"dharshan",java:73,cn:45,os:69}])
```

Step 2: Aggregate function $sum

```
> db.s_mark.aggregate([
    {
      $addFields: {
        totalmark: { $sum: ["$java", "$cn", "$os"] }
      }
    }
  ])
```

Result of $sum

```
< {
    _id: 1,
    sname: 'Arun',
    java: 76,
    cn: 73,
    os: 98,
    totalmark: 247
  }
  {
    _id: 2,
    sname: 'banu',
    java: 67,
    cn: 56,
    os: 76,
    totalmark: 199
  }
```

Step 3: Aggregate function $add $multiply $divide

```
> db.s_mark.aggregate([
    {
      $addFields: {
        totalmark: { $add: ["$java", "$cn", "$os"] },
        percentage: {
          $multiply: [
            { $divide: [{ $add: ["$java", "$cn", "$os"] }, 300] },
            100
          ]
        }
      }
    }
  ])
```

Result of $add $multiply $divide

```
< {
    _id: 1,
    sname: 'Arun',
    java: 76,
    cn: 73,
    os: 98,
    totalmark: 247,
    percentage: 82.33333333333334
  }
  {
    _id: 2,
    sname: 'banu',
    java: 67,
    cn: 56,
    os: 76,
    totalmark: 199,
    percentage: 66.33333333333333
  }
```

Step 4: Update document with field edu_status

```
> db.s_mark.updateOne( {_id:1}, { $set: {"edu_status": "regular"}})
> db.s_mark.updateOne( {_id:2}, { $set: {"edu_status": "intern"}})
> db.s_mark.updateOne( {_id:3}, { $set: {"edu_status": "regular"}})
> db.s_mark.updateOne( {_id:4}, { $set: {edu_status: "intern"}})
```

Step 5: Find documents in Table view

s_mark

	_id Int32	sname String	java Int32	cn Int32
1	1	"Arun"	76	73
2	2	"banu"	67	56
3	3	"chitra"	89	67
4	4	"dharshan"	73	45

Find documents in mongoshell

```
> db.s_mark.find()
< {
    _id: 1,
    sname: 'Arun',
    java: 76,
    cn: 73,
    os: 98,
    edu_status: 'regular'
  }
  {
    _id: 2,
    sname: 'banu',
    java: 67,
    cn: 56,
    os: 76,
    edu_status: 'intern'
  }
```

```
{
  _id: 3,
  sname: 'chitra',
  java: 89,
  cn: 67,
  os: 54,
  edu_status: 'regular'
}
{
  _id: 4,
  sname: 'dharshan',
  java: 73,
  cn: 45,
  os: 69,
  edu_status: 'intern'
}
```

Step 6: Aggregate function $match - Filters documents based on specified conditions, similar to the WHERE clause in SQL.

```
> db.s_mark.aggregate([{$match: {edu_status:"regular"}}])
< {
    _id: 1,
    sname: 'Arun',
    java: 76,
    cn: 73,
    os: 98,
    edu_status: 'regular'
  }
  {
    _id: 3,
    sname: 'chitra',
    java: 89,
    cn: 67,
    os: 54,
    edu_status: 'regular'
  }
```

Step 7: Aggregate function $match with condition

```
> db.s_mark.aggregate([{ $match: { edu_status: "intern", os: { $gte: 70 } } }])
< {
    _id: 2,
    sname: 'banu',
    java: 67,
    cn: 56,
    os: 76,
    edu_status: 'intern'
  }
```

Step 8: Without aggregate function document filtered using find()

```
> db.s_mark.find({edu_status:"intern"})
< {
    _id: 2,
    sname: 'banu',
    java: 67,
    cn: 56,
    os: 76,
    edu_status: 'intern'
  }
  {
    _id: 4,
    sname: 'dharshan',
    java: 73,
    cn: 45,
    os: 69,
    edu_status: 'intern'
  }
```

Step 09: Aggregate function $ group

```
> db.s_mark.aggregate([
    {
      $group: {
        _id: "$edu_status",          // group key
        avg_java: { $avg: "$java" },
        avg_cn: { $avg: "$cn" },
        avg_os: { $avg: "$os" },
        count: { $sum: 1 }           // how many students in each group
      }
    }
  ])
```

Result of $ group

```
{
  _id: 'intern',
  avg_java: 70,
  avg_cn: 50.5,
  avg_os: 72.5,
  count: 2
}
{
  _id: 'regular',
  avg_java: 82.5,
  avg_cn: 70,
  avg_os: 76,
  count: 2
}
```

Step 10: $Count

```
db.s_mark.aggregate([
  {
    $match: { "edu_status": "regular" }
  },
  {
    $count: "totalregularstudents"
  }
])
{
  totalregularstudents: 2
}
```

```
db.s_mark.aggregate([
  {
    $match: { "edu_status": "intern" }
  },
  {
    $count: "Total_Intern_Students"
  }
])
{
  Total_Intern_Students: 2
}
```

```
> use e_shopping
< switched to db e_shopping
> db.s_mark.estimatedDocumentCount()
< 4
> db.s_mark.countDocuments({})
< 4
> db.s_mark.countDocuments({edu_status:"intern"})
< 2
e_shopping>
```

Step 11: Including and Excluding fields using $project and $limit

```
> db.product.aggregate([
    {
      $project: {
        "_id":0,
       "name": 1,
        "category": 1,
        "stock": 1
      }
    },
    {
      $limit: 2
    }
  ])
```

Result of $project

```
< {
    name: 'Laptop',
    category: 'Electronics',
    stock: 50
  }
  {
    name: 'Mouse',
    category: 'Electronics',
    stock: 200
  }
e_shopping>
```

```
> db.student.aggregate([{ $project: {"_id":0,}}])
< {
    firstname: 'adhi',
    lastname: 'ravi',
    age: 16,
    email: 'adhiravi@gmail.com'
  }
  {
    firstname: 'adhi',
    lastname: 'mani',
    age: 12,
    email: 'dhivya@gmail.com'
  }
  {
    firstname: 'balu',
    lastname: 'velu',
    age: 15,
    email: 'baluv@gmail.com'
  }
```

← Excluding "_id" field

Error because MongoDB does not allow mixing inclusion and exclusion in the same $project except for the _id field.

```
> db.student.aggregate([{ $project: {"_id":0,"firstname":1,"lastname":1,"age":0,"email":1}}])
MongoServerError[Location31254]: Invalid $project :: caused by :: Cannot do exclusion on field age in inclusion projection
```

Only Exclusions

```
> db.student.aggregate([
    {
      $project: {
        _id: 0,
        age: 0
      }
    }
  ])
< {
    firstname: 'adhi',
    lastname: 'ravi',
    email: 'adhiravi@gmail.com'
  }
  {
    firstname: 'adhi',
    lastname: 'mani',
    email: 'dhivya@gmail.com'
  }
```

Adding new fields using $project:

```
> db.student.aggregate([{ $project: { "fullName": { $concat: ["$firstname", " ", "$lastname"] } } }])
< {
    _id: 1,
    fullName: 'adhi ravi'
  }
  {
    _id: 2,
    fullName: 'adhi mani'
  }
  {
    _id: 3,
    fullName: 'balu velu'
  }
```

****

******Program 8: Sorting, filtering and Index in MongoDB*****

Step 1 : Use e_shopping database → insert values to product collection

```
> db.product.insertMany([
    { name: "Laptop", category: "Electronics", price: 1200, stock: 50 },
    { name: "Mouse", category: "Electronics", price: 25, stock: 200 },
    { name: "Keyboard", category: "Electronics", price: 75, stock: 150 },
    { name: "Desk Chair", category: "Furniture", price: 300, stock: 30 },
    { name: "Coffee Table", category: "Furniture", price: 150, stock: 40 },
    { name: "Bookcase", category: "Furniture", price: 200, stock: 20 },
    { name: "T-Shirt", category: "Apparel", price: 20, stock: 500 },
    { name: "Jeans", category: "Apparel", price: 60, stock: 100 }
  ]);
```

Step 2: Filter the document based on category wise and price wise

```
> db.product.find({category:"Electronics", price: { $lt: 100 }})
< {
    _id: ObjectId('68f8983adabcf8634ae16e10'),
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    stock: 200
  }
  {
    _id: ObjectId('68f8983adabcf8634ae16e11'),
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    stock: 150
  }
```

Step 3: Sort the document based on price field in ascending order

```
> db.product.find().sort({ price: 1 })
```

Step 4: Filter the document based category: furniture and apply descending order based on stock field

```
> db.product.find({ category: "Furniture" }).sort({ stock: -1 });
```

Step 5: Filter and sort the document based on category, price and name

```
> db.product.find({ category: "Electronics", price: { $gt: 50 } }).sort({ name: 1 });
< {
    _id: ObjectId('68f8983adabcf8634ae16e11'),
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    stock: 150
  }
  {
    _id: ObjectId('68f8983adabcf8634ae16e0f'),
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    stock: 50
  }
```

Step 6: Before applying Index

```
> db.product.getIndexKeys()
< [ { _id: 1 } ]
> db.product.getIndexes()
< [ { v: 2, key: { _id: 1 }, name: '_id_' } ]
```

Step 7: After applying Index

```
> db.product.createIndex({ category: 1, price: 1 });
< category_1_price_1
> db.product.getIndexKeys()
< [ { _id: 1 }, { category: 1, price: 1 } ]
> db.product.getIndexes()
< [
    { v: 2, key: { _id: 1 }, name: '_id_' },
    { v: 2, key: { category: 1, price: 1 }, name: 'category_1_price_1' }
  ]
```

****

****

Program:9 Create a Collection and Document and perform the following:

a) Find a document by id

b) Find a user by email

c) Find a list of all users with the same firstname

d) Find all users who are more than 12 years old

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Find all users who are more than 12 years old

```
> db.student.find({age: {$gt:12}})
< {
    _id: 1,
    firstname: 'adhi',
    lastname: 'ravi',
    age: 16,
    email: 'adhiravi@gmail.com'
  }
  {
    _id: 3,
    firstname: 'balu',
    lastname: 'velu',
    age: 15,
    email: 'baluv@gmail.com'
  }
```

Find a list of all users with the same firstname

```
> db.student.aggregate([
    {
      $group: {
        _id: "$firstname",       // group by firstName field
        count: { $sum: 1 },      // count how many users have that name
        users: { $push: "$$ROOT" } // keep the full user documents
      }
    },
    {
      $match: {
        count: { $gt: 1 }        // only include names that appear more than once
      }
    }
  ]);
```

```
< {
    _id: 'adhi',
    count: 2,
    users: [
      {
        _id: 1,
        firstname: 'adhi',
        lastname: 'ravi',
        age: 16,
        email: 'adhiravi@gmail.com'
      },
      {
        _id: 2,
        firstname: 'adhi',
        lastname: 'mani',
        age: 12,
        email: 'dhivya@gmail.com'
      }
    ]
```

Find a user by email

```
> db.student.findOne(
    { email: { $regex: "@gmail\\.com$", $options: "i" } }
  );
< {
    _id: 1,
    firstname: 'adhi',
    lastname: 'ravi',
    age: 16,
    email: 'adhiravi@gmail.com'
  }
```

```
> db.product.findOne(
    { email: { $regex: "@kgisliim\\.ac\\.in$", $options: "i" } }
  );
< {
    _id: 2,
    username: 'Shruthi',
    age: 19,
    email: 'shruthi@kgisliim.ac.in'
  }
```

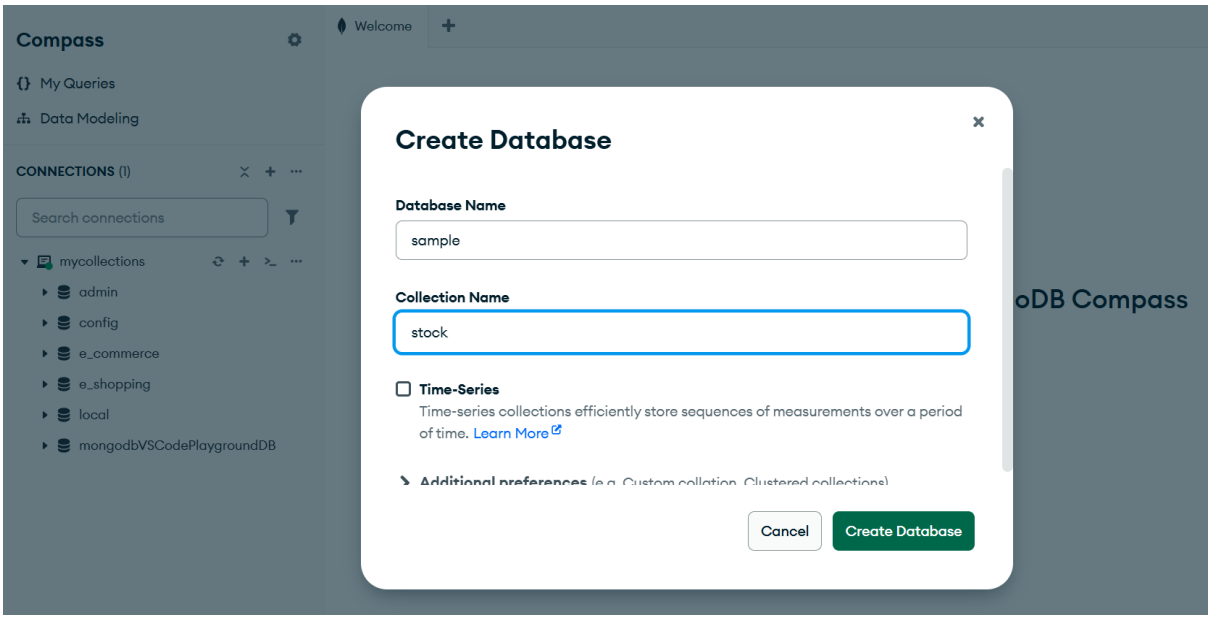
Find a document by id

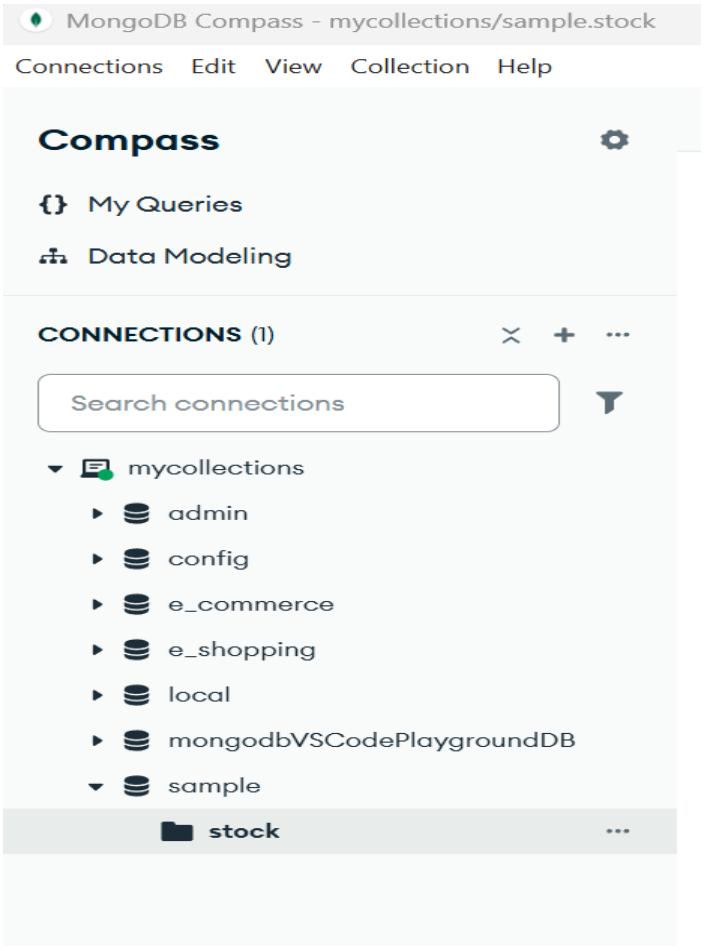
```
> db.student.findOne({ _id:2 });
< {
    _id: 2,
    firstname: 'adhi',
    lastname: 'mani',
    age: 12,
    email: 'dhivya@gmail.com'
  }
```

****

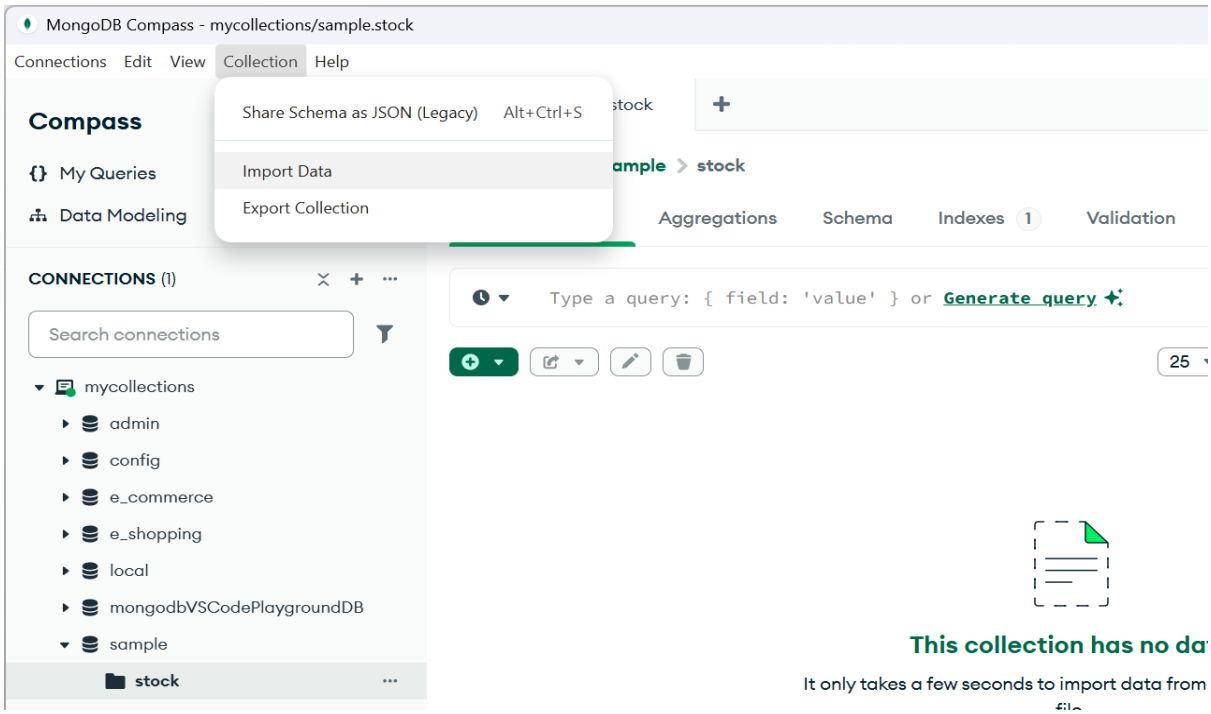
******* Program 10: Use real time data set of Stock exchange by import CSV data*******

Step 1: Create new database sample and collection named as stock

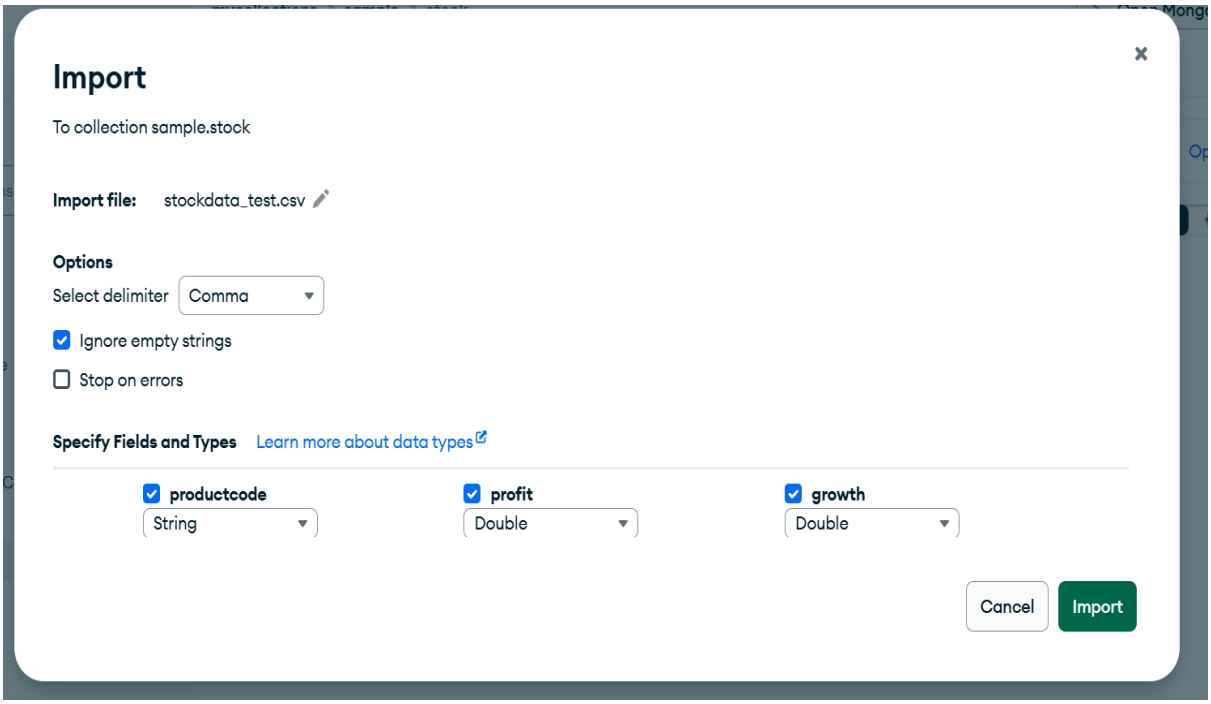




Step 2: Choose Collection menu form menubar and select Import data



Step 3: Choose stock_data.csv from open dialog



Step 4: Imported data in table view

stock

	_id ObjectId	productcode String	profit Double	growth Double
1	ObjectId('68fef252298700e…	"stk001"	0.72	0.5
2	ObjectId('68fef252298700e…	"stk002"	0.45	-0.04
3	ObjectId('68fef252298700e…	"stk003"	1.25	0.32
4	ObjectId('68fef252298700e…	"stk004"	-0.12	-0.06
5	ObjectId('68fef252298700e…	"stk005"	1.05	0.23
6	ObjectId('68fef252298700e…	"stk006"	0.67	0.4

Step 5: Find all the stocks where the profit is over 0.5

```
> db.stock_data.find({
    profit: { $gt: 0.5 }
  });
< {
    _id: ObjectId('68f9045717f0a1ecb33f0d5e'),
    productcode: 'stk001',
    profit: 0.72,
    growth: 0.15
  }
  {
    _id: ObjectId('68f9045717f0a1ecb33f0d60'),
    productcode: 'stk003',
    profit: 1.25,
    growth: 0.32
  }
  {
    _id: ObjectId('68f9045717f0a1ecb33f0d62'),
    productcode: 'stk005',
    profit: 1.05,
    growth: 0.23
  }
  {
    _id: ObjectId('68f9045717f0a1ecb33f0d63'),
    productcode: 'stk006',
    profit: 0.67,
    growth: 0.4
  }
sample >
```

Step 6: Find all the stocks with negative growth

```
> use sample
< switched to db sample
> db.stock_data.find({
    growth: { $lt: 0 }
  });
< {
    _id: ObjectId('68f9045717f0a1ecb33f0d5f'),
    productcode: 'stk002',
    profit: 0.45,
    growth: -0.04
  }
  {
    _id: ObjectId('68f9045717f0a1ecb33f0d61'),
    productcode: 'stk004',
    profit: -0.12,
    growth: -0.06
  }
```

Command line Import method

Import CSV file

```
C:\Program Files\MongoDB\Tools\100\bin>mongoimport --db sample --collection stock --type csv --file D:\ADBMS\Practical\stockdata_test
.csv --headerline
2025-10-27T13:39:43.136+0530    connected to: mongodb://localhost/
2025-10-27T13:39:43.140+0530    6 document(s) imported successfully. 0 document(s) failed to import.
```

Import JSON file

```
C:\Program Files\MongoDB\Tools\100\bin>mongoimport --db sample --collection stock --file "D:\ADBMS\Practical\output_stockdata1.json"
2025-10-27T14:00:00.404+0530    connected to: mongodb://localhost/
2025-10-27T14:00:00.408+0530    6 document(s) imported successfully. 0 document(s) failed to import.
```

Export JSON file format

```
C:\Program Files\MongoDB\Tools\100\bin>mongoexport --db=sample --collection=stock --out="D:\ADBMS\Practical\output_stockdata.json"
2025-10-27T13:45:30.285+0530    connected to: mongodb://localhost/
2025-10-27T13:45:30.292+0530    exported 6 records

C:\Program Files\MongoDB\Tools\100\bin>mongoexport --db sample --collection stock --out "D:\ADBMS\Practical\output_stockdata1.json"
2025-10-27T13:47:34.767+0530    connected to: mongodb://localhost/
2025-10-27T13:47:34.772+0530    exported 6 records
```

Export CSV file format

```
C:\Program Files\MongoDB\Tools\100\bin>mongoexport --db sample --collection stock --out "D:\ADBMS\Practical\output_stock.csv"
2025-10-27T13:53:47.054+0530    connected to: mongodb://localhost/
2025-10-27T13:53:47.061+0530    exported 6 records
```

****